

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
NETWORK SECURITY (CYS360)
(6TH SEMESTER, 3RD Year) LAB EXPERIMENT # 6

Name:	__Muhammad_Usman__		Roll number:	__23BSCYS006__	
Score:	_____	Date:	_____	Instructor's Signature:	_____

Lab # 6: To Access the Internet Using a Static IPv4 Address in Firewall

Objective: The objective of this lab is to access the Internet Using a Static IPv4 Address in Firewall

Duration: 3 Hours

Expected Outcomes: By the end of this lab session, students will be able to:

1. Understand the basic principles of dhcp
2. Understand the basic principles of static routing

Rubrics

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT
SCORE			

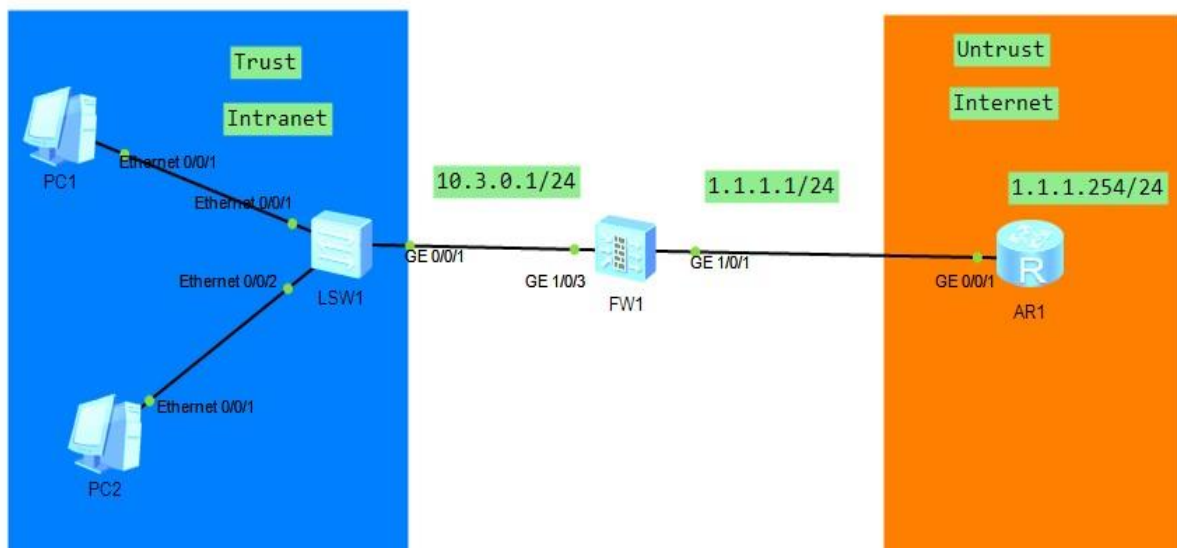
Note: Student performance will be assessed based on this rubric during lab sessions.

Scenario

An enterprise deploys a FW as a security gateway on the network shown in Figure 1 and purchases broadband services from an ISP.

The networking requirements are as follows:

- Intranet PCs communicate with each other using addresses on the network segment 10.3.0.0/24. The FW allocates private network addresses and a DNS server address to the PCs.
- Intranet PCs are able to access the Internet.



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Figure 1 Ethernet link connecting intranet PCs to the Internet

What is DHCP?

DHCP is a **network protocol** that automatically assigns **IP addresses, subnet masks, default gateways, and DNS servers** to devices (like laptops, PCs, or mobile phones) on a network.

Without DHCP, you would have to configure the IP settings manually for every device.

Purpose of DHCP

- **Automatic IP assignment** – no need for manual configuration.
- **Avoids IP conflicts** – ensures that two devices don't get the same IP.
- **Easy management** – especially useful in large networks with many devices.

How DHCP Works (DORA Process)

DHCP works in **four steps**, known as the **DORA process**:

1. **Discover** → The client broadcasts a request saying, *"I need an IP address."*
2. **Offer** → The DHCP server replies with an offered IP (e.g., 192.168.1.10).
3. **Request** → The client asks to use that specific IP.
4. **Acknowledge** → The server confirms and leases the IP to the client.

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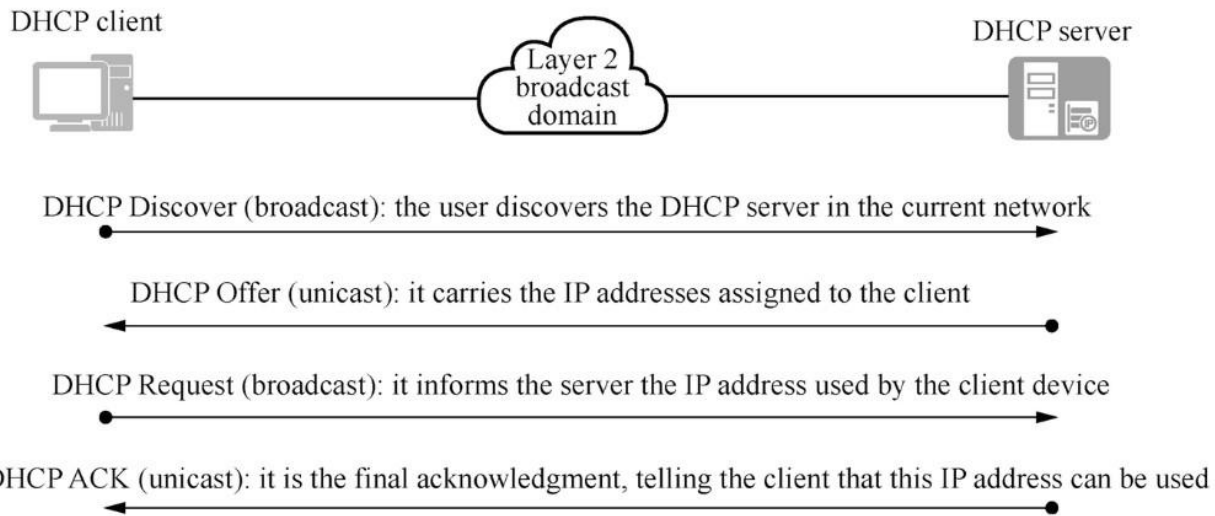


Table 1: Parameters provided by an ISP

Item	Data	Description
Enterprise address	1.1.1.1/24	Public network address that the ISP assigns to the enterprise
Default gateway address	1.1.1.254	Provided by the ISP
DNS server address	9.9.9.9	Provided by the ISP

Configuration Roadmap

The configuration roadmap is as follows:

1. Assign IP addresses to interfaces and add the interfaces to security zones. Set the default gateway address to 1.1.1.254 for GigabitEthernet 1/0/1.
2. Configure the DHCP server function on the FW to allocate IP addresses and a DNS server address to intranet PCs.
3. Configure security policies to allow PCs to access the Internet.
4. Configure NAT policies for source address translation. As the FW translates private addresses into a fixed public network address that is assigned by the ISP, easy-IP is used to simplify the configuration.

Procedure

1. Set the IP addresses of interfaces, and then assign the interfaces to security zones.

```
<FW> system-view
[FW] interface GigabitEthernet 1/0/1
```

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```
[FW-GigabitEthernet1/0/1] ip address 1.1.1.1 255.255.255.0
[FW-GigabitEthernet1/0/1] quit
[FW] interface GigabitEthernet 1/0/3
[FW-GigabitEthernet1/0/3] ip address 10.3.0.1 255.255.255.0
[FW-GigabitEthernet1/0/3] quit
[FW] firewall zone untrust
[FW-zone-untrust] add interface GigabitEthernet 1/0/1
[FW-zone-untrust] quit
[FW] firewall zone trust
[FW-zone-trust] add interface GigabitEthernet 1/0/3
[FW-zone-trust] quit
```

2. Configure the FW as a DHCPv4 Server.

Enable the DHCP function.

```
[FW] dhcp enable
```

Create an interface address pool and specify the default gateway IP address and the DNS server address for PCs on the intranet.

```
[FW] interface GigabitEthernet 1/0/3
[FW-GigabitEthernet1/0/3] dhcp select interface
[FW-GigabitEthernet1/0/3] dhcp server ip-range 10.3.0.1 10.3.0.254
[FW-GigabitEthernet1/0/3] dhcp server dns-list 9.9.9.9
[FW-GigabitEthernet1/0/3] dhcp server gateway-list 10.3.0.1
[FW-GigabitEthernet1/0/3] quit
```

3. Configure security policies, allowing PCs on the intranet to access the Internet.

```
[FW] security-policy
[FW-security-policy] rule name policy_sec_1
[FW-security-policy-sec_policy_1] source-address 10.3.0.0 mask 255.255.255.0
[FW-security-policy-sec_policy_1] source-zone trust
[FW-security-policy-sec_policy_1] destination-zone untrust
[FW-security-policy-sec_policy_1] action permit
[FW-security-policy-sec_policy_1] quit
[FW-security-policy] quit
```

4. Configure a NAT policy, allowing PCs on the intranet to access the Internet by using the resulting public IP address of translation.

```
[FW] nat-policy
[FW-policy-nat] rule name policy_nat_1
[FW-policy-nat-rule-policy_nat_1] source-address 10.3.0.0 mask 255.255.255.0
[FW-policy-nat-rule-policy_nat_1] source-zone trust
[FW-policy-nat-rule-policy_nat_1] egress-interface GigabitEthernet 1/0/1
[FW-policy-nat-rule-policy_nat_1] action source-nat easy-ip
[FW-policy-nat-rule-policy_nat_1] quit
```

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```
[FW-policy-nat] quit
```

5. Configure the default route whose next hop IP address is 1.1.1.254.

```
[FW] ip route-static 0.0.0.0 0.0.0.0 1.1.1.254
```

Configuration Verification

1. View details about GigabitEthernet 1/0/1 and check whether interface GigabitEthernet 1/0/1 has obtained a public IP address and both the physical and IPv4 states are Up.

```
[FW] display interface GigabitEthernet 1/0/1
```

GigabitEthernet 1/0/1 current state : **UP**

Line protocol current state : **UP**

GigabitEthernet 1/0/1 current firewall zone : untrust

Description : GigabitEthernet 1/0/1 Interface, Route Port

The Maximum Transmit Unit is 1500 bytes, Hold timer is 10(sec)

Internet Address is **1.1.1.1/24**

IP Sending Frames' Format is PKTFMT_ETHNT_2, Hardware address is 0022-a100-a101

Media type is twisted pair, loopback not set, promiscuous mode not set 100Mb/s-speed mode, full-duplex mode, link type is auto negotiation max-bandwidth : 100000 Kbps

Last physical up time : -

Last physical down time : 2015-05-07 20:33:13

Current system time: 2015-05-11 10:08:18

Max input bit rate:528530448 bits/sec at 2015-05-07 12:53:46

Max output bit rate:5280418 bits/sec at 2015-05-07 12:54:26

Max input packet rate:750753 packets/sec at 2015-05-07 22:43:46

Max output packet rate:7843 packets/sec at 2015-05-07 22:53:58

Last 300 seconds input rate 8 bytes/sec, 0 packets/sec

Last 300 seconds output rate 8 bytes/sec, 0 packets/sec

Input: 1149 packets, 99478 bytes

12 unicasts, 4 broadcasts, 1133 multicasts, 0 pauses

0 overruns, 0 runts, 0 jumbos, 0 FCS errors

0 length errors, 0 code errors, 0 align errors

0 fragment errors, 0 giants, 0 jabber errors

0 dribble condition detected, 0 other errors

Output: 1104 packets, 94646 bytes

7 unicasts, 10 broadcasts, 1087 multicasts, 0 pauses

0 underruns, 0 runts, 0 jumbos, 0 FCS errors

0 fragment errors, 0 giants, 0 jabber errors

0 collisions, 0 late collisions

0 ex. collisions, 0 deferred, 0 other errors

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2. Run the **ipconfig/all** command on a PC to verify that the PC has obtained a valid IP address and DNS address. The following example uses a PC running Windows XP. The actual command output may vary.

Ethernet adapter Local Area Connection:

```
Connection-specific DNS Suffix . :  
Description . . . . . : Intel(R) 82574L Gigabit Network Connection  
Physical Address. . . . . : 00-1B-21-B4-0B-35  
Dhcp Enabled. . . . . : Yes  
Autoconfiguration Enabled . . . : Yes  
IP Address. . . . . : 10.3.0.2  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . : 10.3.0.1  
DHCP Server . . . . . : 10.3.0.1  
DNS Servers . . . . . : 9.9.9.9  
Lease Obtained. . . . . : Tuesday, December 6, 2011, 05:58:28 AM  
Lease Expires . . . . . : Friday, December 16, 2011, 05:58:28 AM
```

Task:

1. Ping from PC1 to the internet on the given ip address 1.1.1.254
2. Ping from PC2 to the internet on the given ip address 1.1.1.254
3. Attach Screenshots

Note: Attach all the screenshots of the configuration. The Firewall name should be your roll number like FW1-22BSCYS001. Upload the .topo and .cfg files on MS Teams Assignment.

Questions:

1. What do you mean by static routing and how is it useful in finding unknown networks?

Answer:

Static routing is a routing method in which routes are manually configured by the network administrator. It is useful for finding unknown networks by using a default route, which forwards all unknown traffic to a predefined next-hop router, usually the ISP gateway.

2. What do you mean by DHCP and explain the DORA process?

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Answer:

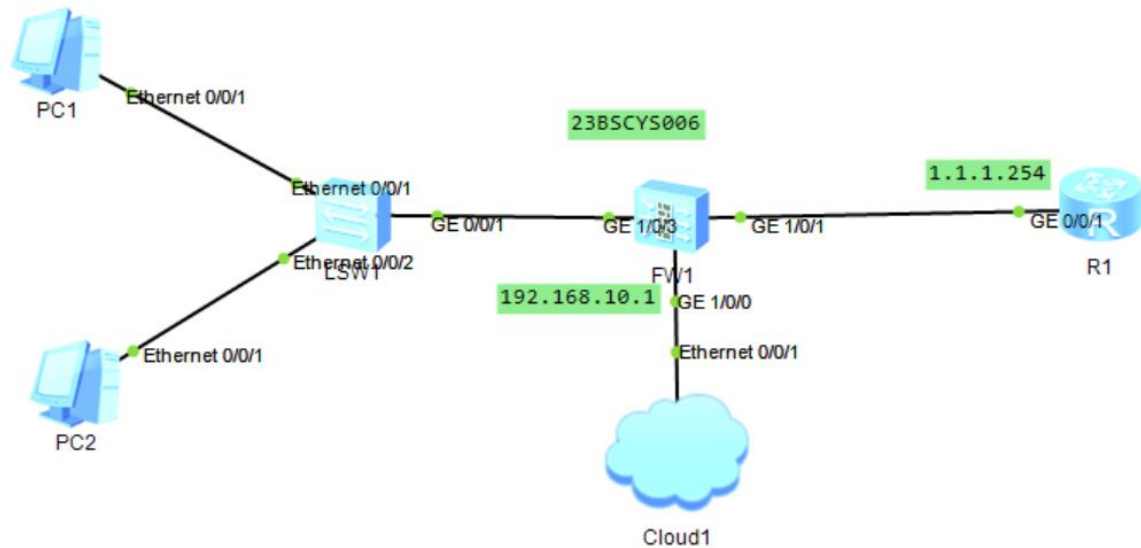
DHCP (Dynamic Host Configuration Protocol) is a network protocol that automatically assigns IP addresses and other network settings to devices. The DORA process includes Discover (client requests an IP), Offer (server offers an IP), Request (client requests the offered IP), and Acknowledge (server confirms and assigns the IP).

3. What is 9.9.9.9 DNS used for?

Answer:

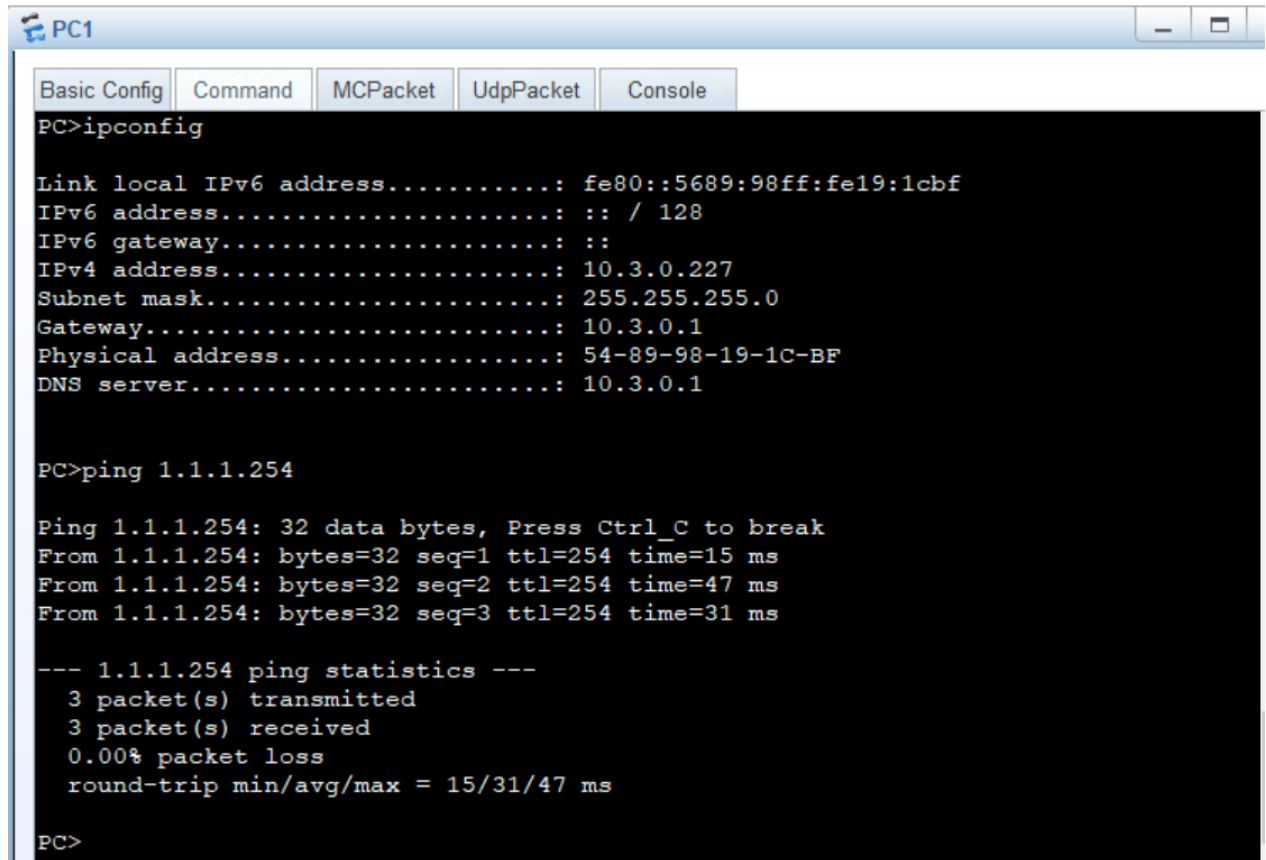
9.9.9.9 is a public DNS server provided by Quad9. It is used to resolve domain names into IP addresses and also helps improve security by blocking access to malicious websites.

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```
CLI Console (connected)
#
return
[FW-1-23BSCYS006-policy-security]q
[FW-1-23BSCYS006]q
^
Error: Ambiguous command found at '^' position.
[FW-1-23BSCYS006]display interface GigabitEthernet 1/0/1
2026-01-26 11:37:13.220 +05:00
GigabitEthernet1/0/1 current state : UP
Line protocol current state : UP
Last line protocol up time : 2026-01-26 11:05:26 UTC+05:00
Description: Huawei, USG6000V1-ENSP Series, GigabitEthernet1/0/1 Interface
Route Port, The Maximum Transmit Unit is 1500
Internet Address is 1.1.1.1/24
IP Sending Frames' Format is PKTFMT_ETHNT_2, Hardware address is 00e0-fc73-1175
Max-bandwidth : 1000000 kbps
Last physical up time : 2026-01-26 05:53:21
Last physical down time : 2026-01-26 05:48:03
Current system time: 2026-01-26 11:37:13+05:00
Top 3 input bit rate: -
```


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The screenshot shows a Windows-style window titled "PC1" with a tabbed interface. The "Command" tab is active, displaying a command prompt. The user has entered the command "ipconfig", which has returned the following network configuration details:

```
Link local IPv6 address.....: fe80::5689:98ff:fe19:1cbf
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 10.3.0.227
Subnet mask.....: 255.255.255.0
Gateway.....: 10.3.0.1
Physical address.....: 54-89-98-19-1C-BF
DNS server.....: 10.3.0.1
```

Next, the user has entered the command "ping 1.1.1.254". The output shows three successful ping attempts:

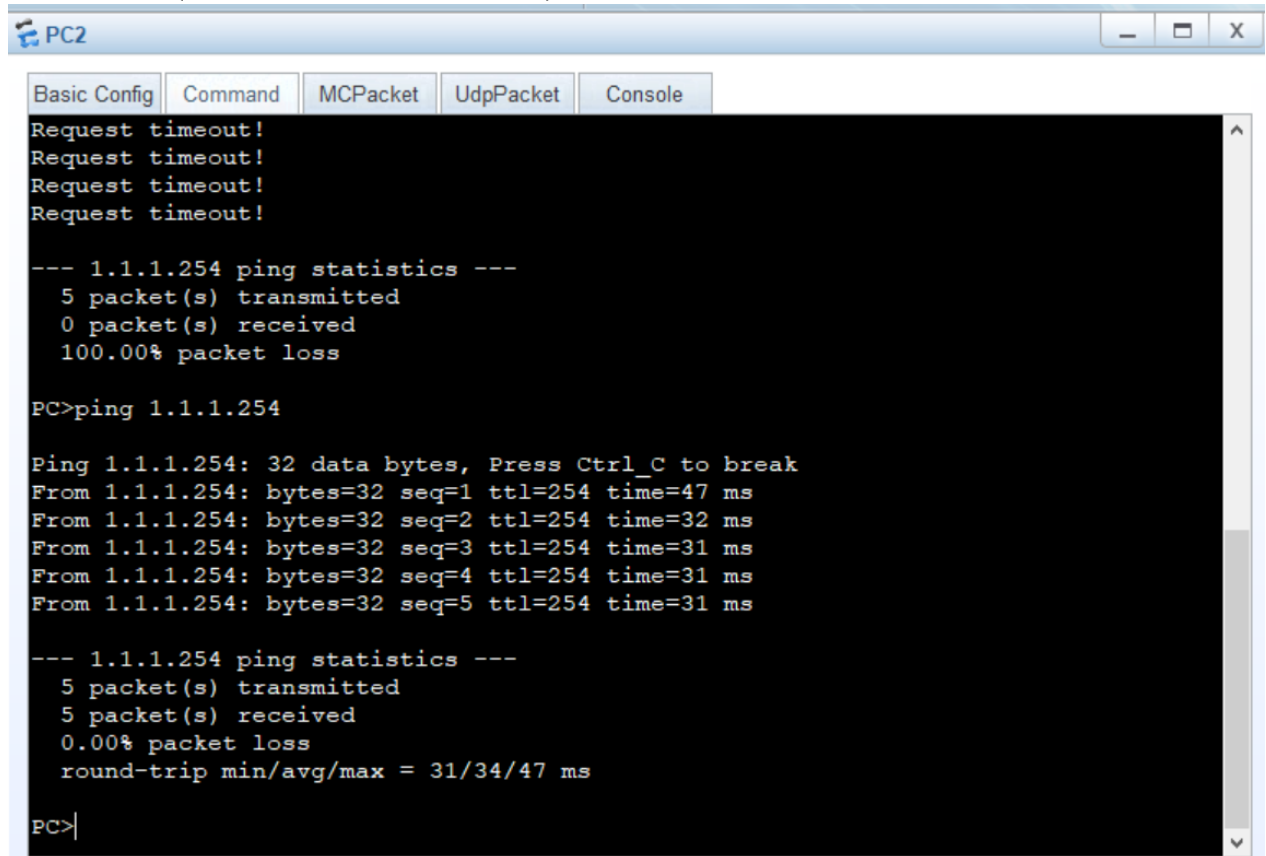
```
Ping 1.1.1.254: 32 data bytes, Press Ctrl_C to break
From 1.1.1.254: bytes=32 seq=1 ttl=254 time=15 ms
From 1.1.1.254: bytes=32 seq=2 ttl=254 time=47 ms
From 1.1.1.254: bytes=32 seq=3 ttl=254 time=31 ms
```

Below the ping results, the statistics are displayed:

```
--- 1.1.1.254 ping statistics ---
 3 packet(s) transmitted
 3 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 15/31/47 ms
```

The command prompt ends with "PC>" on a new line.

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The screenshot shows a PC2 console window with a tabbed interface. The 'Console' tab is active, displaying a series of commands and their outputs. The first part shows four 'Request timeout!' messages. This is followed by a 'ping 1.1.1.254' command that fails, showing 0 packets received and 100% packet loss. Then, the user enters 'PC>ping 1.1.1.254', which shows five successful pings with response times between 31 and 47 ms. Finally, another 'ping 1.1.1.254' command is shown, which succeeds with 5 packets received and 0.00% packet loss. The console ends with 'PC>' and a cursor.

```
PC2
Basic Config | Command | MCPacket | UdpPacket | Console
Request timeout!
Request timeout!
Request timeout!
Request timeout!

--- 1.1.1.254 ping statistics ---
  5 packet(s) transmitted
  0 packet(s) received
 100.00% packet loss

PC>ping 1.1.1.254

Ping 1.1.1.254: 32 data bytes, Press Ctrl_C to break
From 1.1.1.254: bytes=32 seq=1 ttl=254 time=47 ms
From 1.1.1.254: bytes=32 seq=2 ttl=254 time=32 ms
From 1.1.1.254: bytes=32 seq=3 ttl=254 time=31 ms
From 1.1.1.254: bytes=32 seq=4 ttl=254 time=31 ms
From 1.1.1.254: bytes=32 seq=5 ttl=254 time=31 ms

--- 1.1.1.254 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
 round-trip min/avg/max = 31/34/47 ms

PC>
```